

□ RESPIRATORY SYSTEM — FULL MIND MAP

Biology Ch. 7.iv · Human Anatomy & Physiology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Mammalian Respiration & Lungs
2. Inhaled vs Exhaled Air — Nitrogen, Oxygen, CO₂ & Water Vapour
3. Oxygen Transport — Lungs → Blood → Tissues
4. Alveoli — Site of Gaseous Exchange
5. Low Blood Oxygen & Breathing Rate
6. Carbon Monoxide Poisoning & Haemoglobin
7. ATP — Energy Currency & ATP → ADP
8. Aerobic Respiration & Glucose Breakdown
9. Pyruvate — Aerobic / Anaerobic Pathways
10. Source-Specific Body Temperature Question

1. Mammalian Respiration & Lungs

- Mammals mein respiration ka primary organ lungs (फेफड़े) hain. Source ke mutabik two lungs heart ke dono sides, backbone ke paas located hote hain; ye atmosphere se oxygen blood mein transfer aur blood se carbon dioxide atmosphere mein release karne ke gas-exchange process mein kaam karte hain. [Q1]

2. Inhaled vs Exhaled Air — Nitrogen, Oxygen, CO₂ & Water Vapour

- Respiration mein inhaled aur exhaled air compare karne par source ke according nitrogen ki amount unchanged rehti hai. Expired air mein oxygen percentage decrease hota hai, jabki carbon dioxide aur water vapour ki amount increase hoti hai. [Q2]

3. Oxygen Transport — Lungs → Blood → Tissues

- Body cells tak oxygen pahunchne ka correct sequence source mein: air se Lungs → Blood → Tissue/cells. Pehle lungs oxygen uptake karte hain, phir blood us oxygen ko transport karta hai, aur finally oxygen tissues/cells tak deliver hoti hai. [Q3]

4. Alveoli — Site of Gaseous Exchange

- Lungs mein gaseous exchange ka main site alveoli hai. Alveoli ko capillaries surround karti hain; inhaled oxygen alveoli se capillary blood mein move karti hai aur carbon dioxide capillary blood se alveolar air mein move karti hai. [Q4]

5. Low Blood Oxygen & Breathing Rate

- Blood oxygen concentration kam hone par breathing rate increase hoti hai. Source is low blood-oxygen condition ko hypoxemia ke context mein explain karta hai aur shortness of breath se link karta hai. [Q5]

6. Carbon Monoxide Poisoning & Oxygen-Carrying Capacity

- Carbon monoxide haemoglobin ke saath combine karke carboxyhaemoglobin (HbCO) banata hai. Isse haemoglobin tissues tak oxygen effectively carry nahi kar pata aur blood ki oxygen-carrying capacity reduce hoti hai, leading to hypoxia. Source HbCO complex ko fairly stable bhi batata hai. [Q6]

7. ATP — Energy Currency & ATP → ADP

- ATP (Adenosine Triphosphate) high-energy molecule hai. Source ke according maximum energy-providing biotransformation ATP → ADP hai; ATP hydrolysis life processes jaise muscle contraction, membrane gradients aur biosynthetic work ke liye usable energy release karti hai: $\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{Pi} + \text{energy}$. [Q7]

- Cellular respiration nutrients ki biochemical energy ko ATP mein convert karti hai; isliye respiration ke dauran energy ATP ke form mein produced/stored hoti hai. [Q8]

8. Aerobic Respiration — Complete Oxidation of Glucose

- Oxygen ki presence mein glucose ka complete conversion carbon dioxide + water ke saath energy release kare to ise aerobic respiration kehte hain. Source ke mutabik is process mein large amount of energy release hoti hai aur ye plants aur animals ki cells mein hota hai. [Q9]

9. Pyruvate — Aerobic / Anaerobic Pathways

- Glucose cytoplasm mein glycolysis se pyruvate mein break hota hai. Source clearly batata hai ki pyruvate 3-carbon molecule hai, 4-carbon nahi. Oxygen present ho to pyruvate se $\text{CO}_2 + \text{H}_2\text{O} + \text{energy}$ banti hai; oxygen ki kami mein muscles mein lactic acid banta hai; oxygen absent hone par source pyruvate ko $\text{CO}_2 + \text{ethanol}$ mein convert hota batata hai. [Q10]

EXAM TRAP — Q10 Statement Combination

- ▶ Statement (i) “pyruvate has four carbon molecules” false hai because source pyruvate ko 3-carbon molecule batata hai. Statements (ii) lactic-acid formation in oxygen shortage aur (iii) anaerobic $\text{CO}_2 + \text{ethanol}$ formation ko true maana gaya hai; hence source answer (C) = (ii) and (iii). [Q10]

10. Source-Specific Body Temperature Question

- Q11 mein source answer key “Lungs” ko body temperature maintain karne wala part select karti hai. Explanation breathing, oxidation-generated energy aur body-water vaporisation ke through temperature maintenance describe karti hai. Is source-specific framing ko without outside correction preserve kiya gaya hai. [Q11]